Full Stack Development with MERN

Project Documentation

1. Introduction

- **Project Title:** Online Complaint Registration
- **Team Members:** Aakula Srija (Individual Project)

2. Project Overview

• **Purpose:** The Online Complaint Registration System is developed to provide a simple and efficient platform for users to submit, track, and resolve complaints online. It improves transparency and communication between users and administrators.

• **Features:** The system allows users to register, log in, submit complaints, track complaint status, and provide feedback after resolution. Administrators can monitor complaints, update statuses, and manage the overall complaint resolution process through a dedicated dashboard.

3. Architecture

- **Frontend:** React.js is used to create a responsive and interactive user interface for complaint management.
- **Backend:**Node.js and Express.js are used to build secure RESTful APIs and handle application logic. .
- **Database:** MongoDB stores user, complaint, and feedback data using Mongoose schemas for efficient data management

4. Setup Instructions

- **Prerequisites:** Node.js, MongoDB, npm, and a code editor such as Visual Studio Code are required to run the project.

- **Installation:** Clone the repository, install dependencies using `npm install`, configure the `.env` file, and start the frontend and backend servers.

5. Folder Structure

- **Client:** React.js frontend containing components, pages, services, and routing for the user interface.

• **Server:** Node.js and Express.js backend containing routes, controllers, models, middleware, and configuration files.

6. Running the Application

- **Frontend:** `npm run dev` in the frontend directory.
- **Backend:** `npm start` in the server directory.

7. API Documentation

The backend exposes RESTful APIs for user authentication,

complaint management, status updates, and feedback submission

using HTTP methods such as GET, POST, and PUT.

8. Authentication

• Authentication is implemented using JWT (JSON Web Tokens), where users receive a token after login and use it to access protected routes securely.

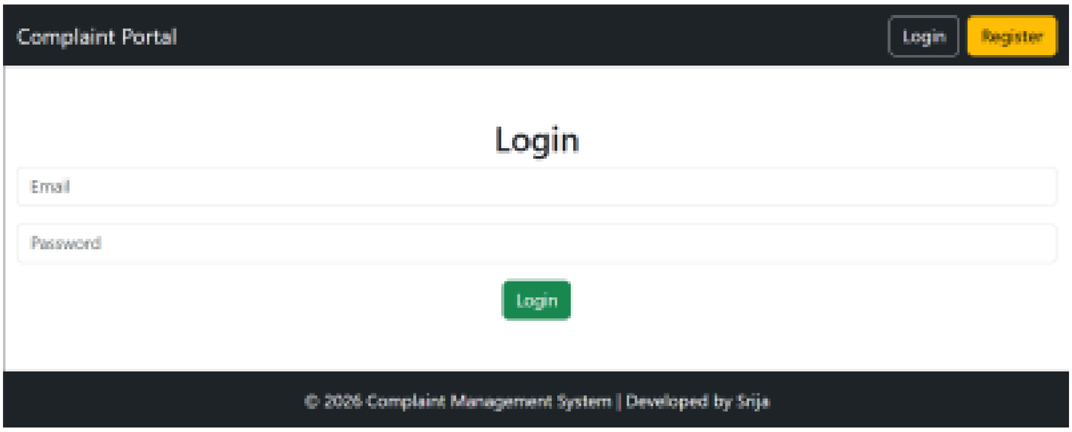
9. User Interface

• The user interface consists of multiple pages, including the Home Page, User Registration Page, Login Page, User Dashboard, Complaint Submission Form, Complaint Tracking Page, Feedback Section, and Admin Dashboard. Screenshots of these pages are included below to demonstrate the functionality and user experience of the application.

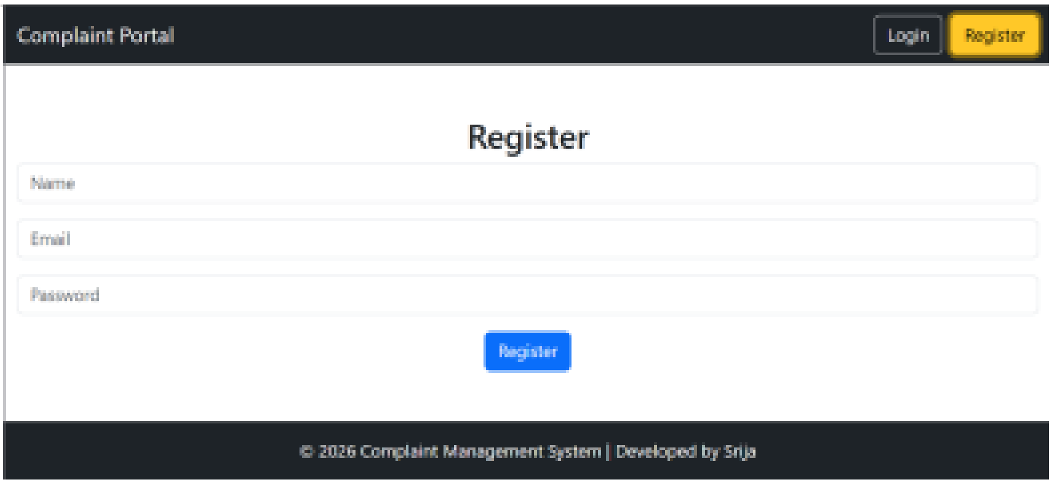
- Home Page
- Login Page
- Registration Page
- User Dashboard
- Complaint Submission
- Complaint Tracking
- Feedback System
- Admin Dashboard

Home page

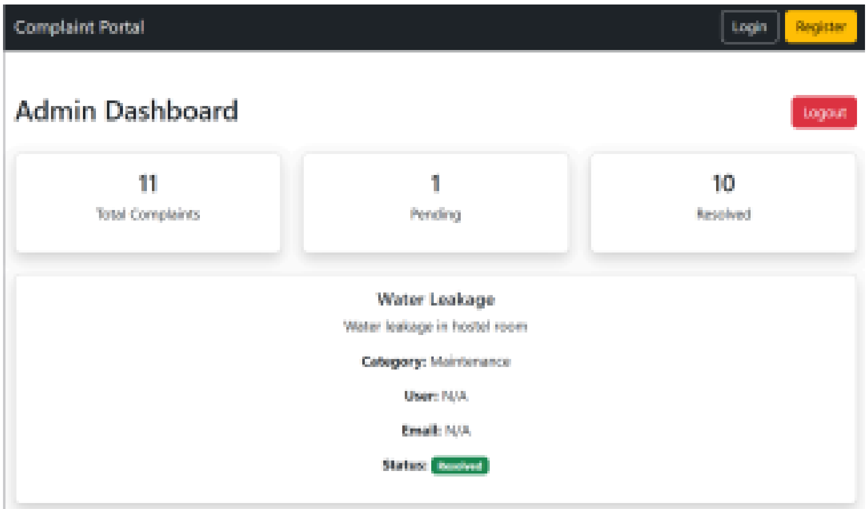
Login Page



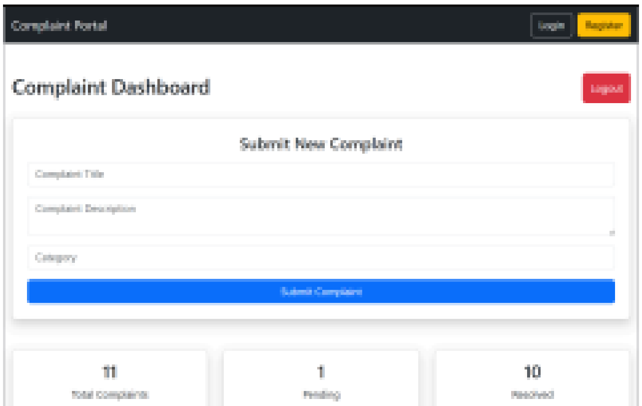
Register Page



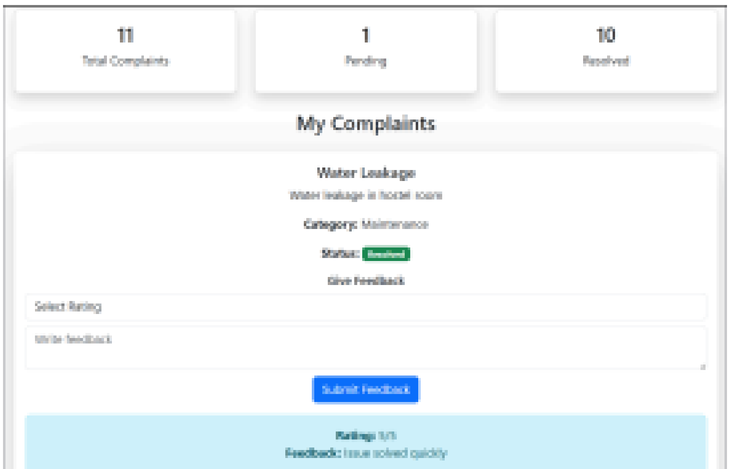
Admin Dashboard



User Dashboard



FeedBack Page



10. Testing

• The application was tested using Thunder Client/Postman to verify all API endpoints and responses.

Manual testing was performed to validate user registration, login, complaint submission, tracking, and feedback features.

The system was also tested to ensure proper authentication, authorization, and smooth user experience across different modules.

11. Screenshots or Demo

The application was successfully developed and tested in a local environment using React.js, Node.js, Express.js, and MongoDB. Screenshots of the major modules, including User Registration, Login, Complaint Submission, Complaint Tracking, Feedback System, and Admin Dashboard, are provided in Section 9.

The demo showcases the complete workflow of the system, from user authentication and complaint registration to complaint resolution and feedback submission. All core functionalities were tested and verified to ensure smooth operation and a user-friendly experience.

GitHub Repository:
https://github.com/akulasrija38-svg/VIP-C2-Online-Complaint-System

12. Known Issues

• The system currently supports basic complaint management and may require additional enhancements for advanced features such as email notifications and agent assignment.

Minor UI improvements and performance optimizations can be implemented in future versions.

No major bugs were identified during testing, and all core functionalities are working as expected.

13. Future Enhancements

• Future enhancements may include email notifications for complaint updates, agent assignment and management, and advanced complaint search and filtering options. The system can also be extended with real-time chat, analytics dashboards, and mobile application support to improve user experience and efficiency.